Activity Series and Solubility Rules

Honors Chemistry reference sheet

Activity Series of the Metals

Most active at the top. Each metal is shown with the ion it forms.

Metal		Forms	Replaces hydrogen?
Lithium	**Li**	$Li^+ + e^-$	**React with cold water, steam and acids**
Cesium	**Cs**	$Cs^+ + e^-$	
Rubidium	**Rb**	$Rb^+ + e^-$	
Potassium	**K**	$K^+ + e^-$	
Barium	**Ba**	$Ba^{2+} + 2\ e^-$	
Strontium	**Sr**	$Sr^{2+} + 2\ e^-$	
Calcium	**Ca**	$Ca^{2+} + 2\ e^-$	
Sodium	**Na**	$Na^+ + e^-$	
Magnesium	**Mg**	$Mg^{2+} + 2\ e^-$	**React with steam and acids**
Aluminum	**Al**	$Al^{3+} + 3\ e^-$	
Manganese	**Mn**	$Mn^{2+} + 2\ e^-$	
Zinc	**Zn**	$Zn^{2+} + 2\ e^-$	
Chromium	**Cr**	$Cr^{3+} + 3\ e^-$	
Iron	**Fe**	$Fe^{2+} + 2\ e^-$	
Cadmium	**Cd**	$Cd^{2+} + 2\ e^-$	
Cobalt	**Co**	$Co^{2+} + 2\ e^-$	**React with acids**
Nickel	**Ni**	$Ni^{2+} + 2\ e^-$	
Tin	**Sn**	$Sn^{2+} + 2\ e^-$	
Lead	**Pb**	$Pb^{2+} + 2\ e^-$	
Hydrogen	**H_2**	**$2\ H^+ + 2\ e^-$**	***reference***
Copper	**Cu**	$Cu^{2+} + 2\ e^-$	**Do not replace hydrogen**
Silver	**Ag**	$Ag^+ + e^-$	
Mercury	**Hg**	$Hg^{2+} + 2\ e^-$	
Platinum	**Pt**	$Pt^{2+} + 2\ e^-$	
Gold	**Au**	$Au^{3+} + 3\ e^-$	

Halogens: $F_2 > Cl_2 > Br_2 > I_2$
most active first; a halogen replaces a halide below it

Using the series (single replacement)

- A metal replaces any metal **below** it from a compound. If it is not below, there is **no reaction**.
- A metal above H_2 replaces the hydrogen of an acid; H_2 forms.
- Only the metals from Li to Na replace hydrogen from cold water (HOH); a metal hydroxide and H_2 form.

Solubility Rules for Ionic Compounds in Water

Always soluble: compounds of the **Group 1** ions (Li^+, Na^+, K^+, Rb^+, Cs^+) and of ammonium, NH_4^+.

Compounds of these ions are soluble (aq) ...

Negative ion		... except with (insoluble)	*slightly soluble with*
nitrate	NO_3^-	*none*	
acetate	$C_2H_3O_2^-$	*none*	Ag^+
chlorate	ClO_3^-	*none*	
chloride	Cl^-	Ag^+, Hg_2^{2+}	Pb^{2+}
bromide	Br^-	Ag^+, Hg_2^{2+}	Pb^{2+}
iodide	I^-	Ag^+, Hg_2^{2+}, Pb^{2+}	
sulfate	SO_4^{2-}	Ba^{2+}, Sr^{2+}, Pb^{2+}, Hg_2^{2+}	Ca^{2+}, Ag^+

Compounds of these ions are insoluble (s) ...

Negative ion		... except with (soluble)	*slightly soluble with*
carbonate	CO_3^{2-}	Group 1, NH_4^+	
phosphate	PO_4^{3-}	Group 1, NH_4^+	
hydroxide	OH^-	Group 1, Ba^{2+}	Ca^{2+}, Sr^{2+}
sulfide	S^{2-}	Group 1, NH_4^+, Ca^{2+}, Sr^{2+}, Ba^{2+}	
sulfite	SO_3^{2-}	Group 1, NH_4^+	
chromate	CrO_4^{2-}	Group 1, NH_4^+, Mg^{2+}	Ca^{2+}
silicate	SiO_3^{2-}	Group 1	
oxide	O^{2-}	Group 1, Ca^{2+}, Sr^{2+}, Ba^{2+} *(these oxides react with water and form hydroxides)*	

Using the rules (double replacement)

- Trade the partners of the two compounds and look up **both** products.
- Find the negative ion first, then check whether the positive ion is an exception.
- An insoluble product is a solid, (s): a reaction happens. If both products are soluble, (aq), there is **no reaction**.

- *Slightly soluble* compounds dissolve a little: a solid forms only from concentrated solutions.
- Hg_2^{2+} is the mercury(I) ion. $PbCl_2$ and $PbBr_2$ dissolve in hot water.
- Li_2CO_3 and Li_3PO_4 are only slightly soluble. NH_4OH is not a compound: it is NH_3 dissolved in water.